

- 📖 **Binary**
- 📖 **Boolean Algebra**
- 📖 **Logic Gates**
- 📖 **Computer Arithmetic**

No skipping. No rushing. Just consistency.

📖 **30-DAY TECH FOUNDATION PLAN**

📅 **30 minutes/day**

Goal: **Strong logic + exam confidence**

WEEK 1: Binary System (FOUNDATION)

Day 1

- What is number system?
- Decimal vs Binary
- Why computers use binary

Day 2

- Binary counting (0–15)
- Bits & bytes

Day 3

- Binary → Decimal conversion

Day 4

- Decimal → Binary conversion

Day 5

- Binary addition
- Carry rules

Day 6

- Binary subtraction
- Borrow concept

Day 7

- Revision + 20 MCQs (binary only)
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WEEK 2: Computer Arithmetic (VERY IMPORTANT)

Day 8

- Signed vs unsigned numbers
- Range of numbers

Day 9

- 1's complement
- Why it is used

Day 10

- **2's complement (MOST IMPORTANT)**
- Why computers use it

Day 11

- Addition using 2's complement

Day 12

- Subtraction using 2's complement

Day 13

- Overflow concept
- Errors in arithmetic

Day 14

- Full revision + exam MCQs
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WEEK 3: Boolean Algebra (LOGIC BUILDER)

Day 15

- What is Boolean algebra?
- Boolean variables (0 & 1)

Day 16

- Basic operators: AND, OR, NOT

Day 17

- Boolean laws:
 - Identity
 - Null
 - Idempotent

Day 18

- Complement laws
- Involution law

Day 19

- De Morgan's Theorem (exam favorite)

Day 20

- Simplification of expressions
- Solve 5–10 problems

Day 21

- Revision + MCQs (Boolean)
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WEEK 4: Logic Gates (REAL HARDWARE LOGIC)

Day 22

- What are logic gates?
- AND, OR, NOT (symbols + truth tables)

Day 23

- NAND & NOR gates
- Why they are called **universal gates**

Day 24

- XOR & XNOR gates
- Real-life examples

Day 25

- Truth tables → expressions
- Expressions → gates

Day 26

- Convert using De Morgan's theorem

- Gate-level simplification

Day 27

- Combine gates problems
 - Exam-type questions
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FINAL PRO LEVEL (CONFIDENCE WEEK)

Day 28

- Mixed practice:
 - Binary
 - Boolean
 - Gates

Day 29

- 40–50 MCQs
- Timed practice (like real exam)

Day 30

- Final revision
 - Teach the concepts aloud
 - Weak-area fixing
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HOW TO STUDY EACH DAY (IMPORTANT)

30 minutes =

- ☐ 10 min → Learn concept
 - ✍☐ 10 min → Examples
 - 10 min → MCQs / revision
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GOLDEN RULES (DON'T IGNORE)

- ? Don't memorize formulas blindly
- ? Always ask: *"WHY does this work?"*
- ? Don't jump days
- ? Revise yesterday for 5 mins